

Introduction to Biophysics II: Quantum Biology

Lecture 02 Molecular orbitals. Spin.

- π -orbitals
- HOMO & LUMO
- Slater-type orbitals
- Schrödinger equation in matrix form
- Spin: singlet & triplet states

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Atomic orbitals

Atomic orbitals: solution for 3D Schrödinger equation for Hydrogen-like atom with 1 electron

$$\frac{-\hbar^2}{2m} \left[\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} \right] + U(x, y, z) \Psi(x, y, z) = E \Psi(x, y, z)$$

Solution in spherical coordinates:

$$\psi_{nlm} = R_{n,l}(r) Y_{l,m}(\theta, \phi)$$

$$E_n = -16\pi^2 Z^2 m_r e^4 / n^2 \hbar^2$$

Z – nuclear charge

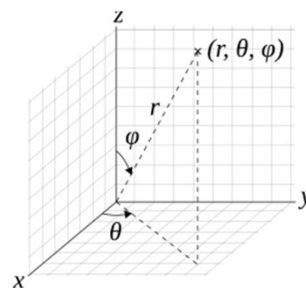
m_r – the reduced mass of electron & nucleus

Quantum numbers:

$n = 1, 2, 3, \dots$ ← Principal quantum number: (defines energy)

$l = 0, 1, 2, \dots, n-1$ Azimuthal quantum number (angular momentum)

$m = -l, \dots, 0, \dots, +l$ Magnetic quantum number



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Atomic orbitals

Atomic orbitals for hydrogen-like atom of charge Z

Table 2.1 Hydrogen-atom wavefunctions

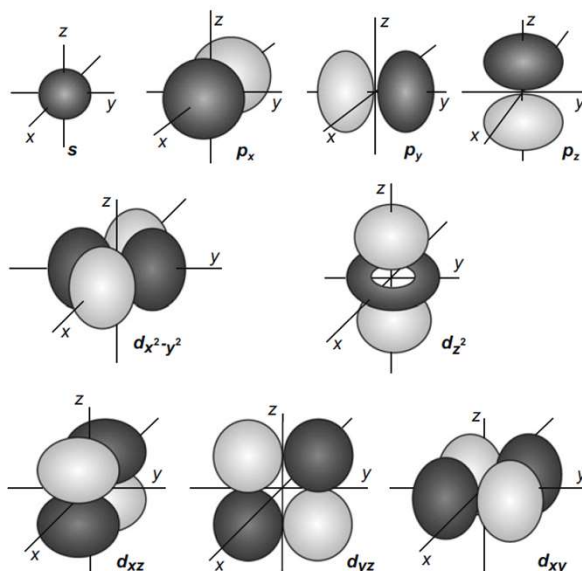
Orbital	n	l	m	$R_{n,l}(r)$ ^a	$Y_{l,m}(\theta, \phi)$ ^b
1s	1	0	0	$(Z/a_0)^{3/2} 2 \exp(-\rho)$	$(2\sqrt{\pi})^{-1}$
2s	2	0	0	$\frac{(Z/a_0)^{3/2}}{2\sqrt{2}} (2 - \rho) \exp(-\rho/2)$	$(2\sqrt{\pi})^{-1}$
$2p_z$	2	1	0	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/4\pi)^{1/2} \cos \theta$
$2p_-$	2	1	-1	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/8\pi)^{1/2} \sin \theta \{ \cos \phi - i \sin \phi \}$ $= (3/8\pi)^{1/2} \sin \theta \exp(-i\phi)$
$2p_+$	2	1	1	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/8\pi)^{1/2} \sin \theta \{ \cos \phi + i \sin \phi \}$ $= (3/8\pi)^{1/2} \sin \theta \exp(i\phi)$

^a $\rho = Zr/a_0$, where Z is the nuclear charge, r is the distance from the nucleus and a_0 is the Bohr radius (0.529 Å)

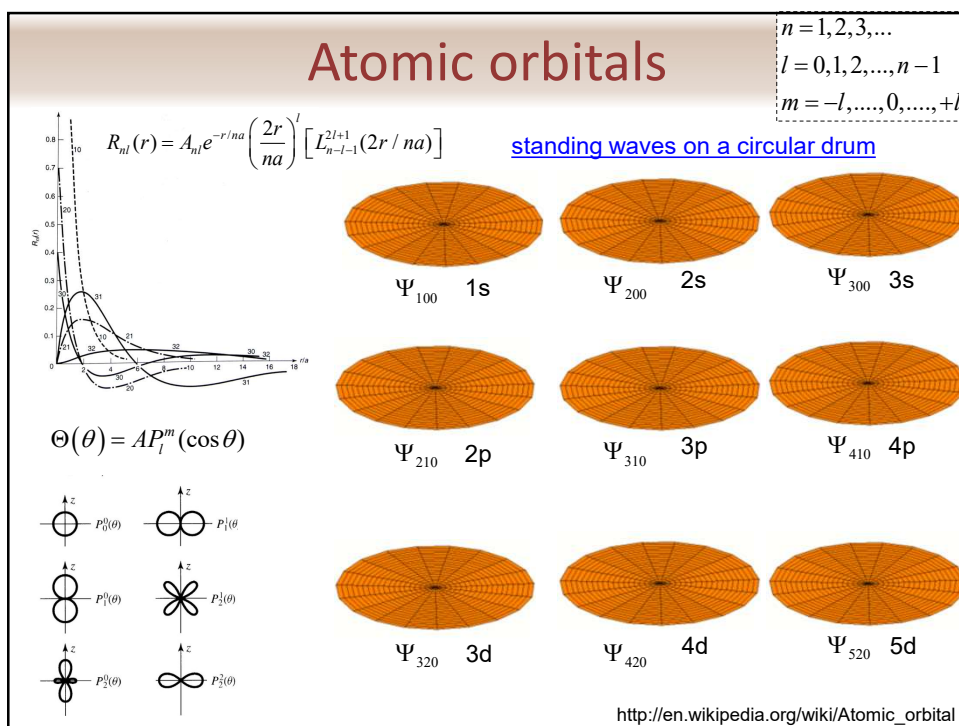
^b θ and ϕ are the angles with respect to the z and x axes in polar coordinates (Fig. 4.4)

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Atomic orbitals



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Molecular orbitals

Multielectron/multiatom orbitals:

Schrödinger Equation can be solved exactly only for H_2^+ ion !
 Any other complex molecule has no exact solution for orbitals...

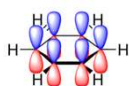
Approximation: represent a molecular electronic wave function by a linear combination of hydrogen atomic orbitals centered at the nuclear positions. In principle, we should include the entire set of atomic orbitals for each nucleus, but smaller sets often provide useful approximations.

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Molecular orbitals

Example: π -orbitals of a molecule with N conjugated atoms can be written as:

Benzene, C_6H_6



$$\psi_k \approx \sum_{n=1}^N C_n^k \psi_{2z(n)}$$

Atomic $2p_z$ orbital centered on atom n

Contribution to molecular orbital k

$$\text{Orthonormalized: } \langle \psi_{2z(n)} | \psi_{2z(n)} \rangle = 1$$

$$\langle \psi_{2z(n)} | \psi_{2z(m)} \rangle = 0 \text{ for } m \neq n$$

$$\text{Molecular orbital should be also normalized: } \sum_{n=1}^N |C_n^k|^2 = 1$$

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Molecular orbitals

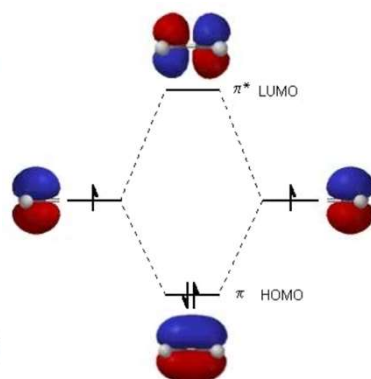
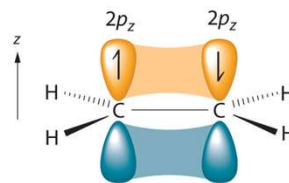
Example: ethylene

HOMO - highest occupied molecular orbital
"bonding" orbital, in-phase,
Constructive interference between carbon atoms
Symmetric combination:

$$\begin{aligned} \psi_{\pi}(\mathbf{r}) &\approx 2^{-1/2} \psi_{2p_z(1)} + 2^{-1/2} \psi_{2p_z(2)} \\ &= 2^{-1/2} \psi_{2p_z}(\mathbf{r} - \mathbf{r}_1) + 2^{-1/2} \psi_{2p_z}(\mathbf{r} - \mathbf{r}_2) \end{aligned}$$

LUMO - Lowest unoccupied molecular
antibonding, out of phase,
destructive interference between atoms,
antisymmetric combination:

$$\begin{aligned} \psi_{\pi^*}(\mathbf{r}) &\approx 2^{-1/2} \psi_{2p_z(1)} - 2^{-1/2} \psi_{2p_z(2)} \\ &= 2^{-1/2} \psi_{2p_z}(\mathbf{r} - \mathbf{r}_1) - 2^{-1/2} \psi_{2p_z}(\mathbf{r} - \mathbf{r}_2) \end{aligned}$$



<http://www.chem.ucalgary.ca/courses/350/Carey5th/Ch10/ch10-6-1.html>

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Molecular orbitals

Complex molecules:

- Techniques for finding the best coefficients C_{ki} for more complex molecules have been refined by comparing calculated molecular energies, dipole moments and other properties with experimental measurements, and a variety of software packages for such calculations are available
- The basis wave functions employed in these descriptions usually are not the atomic orbitals obtained by solving the Schrödinger equation for the hydrogen atom, but rather idealized wave functions with mathematical forms that are easier to manipulate.

Standard form: *Slater-type orbital*

$$\psi_{nlm} = N r^{n^*-1} \exp\{-\zeta r\} Y_{l,m}$$

N – normalization factor

n^* , ζ - parameters related to principal quantum number n

$Y_{l,m}$ - spherical harmonic function of the polar coordinates θ and ϕ

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Molecular orbitals

Slater-type orbital $\psi_{nlm} = N r^{n^*-1} \exp\{-\zeta r\} Y_{l,m}$

N – normalization factor

n^* , ζ - parameters related to principal quantum number n

$Y_{l,m}$ - spherical harmonic function of the polar coordinates θ and ϕ

Example: $2p_z$ orbital for C,N,O: $\psi_{nlm} = (\zeta^5/\pi)^{1/2} r \cdot \cos(\theta) \exp(-\zeta r)$

$$\zeta(\text{C}) = 3.071 \text{ \AA}^{-1} \quad \zeta(\text{N}) = 3.685 \text{ \AA}^{-1} \quad \zeta(\text{O}) = 4.299 \text{ \AA}^{-1}$$

Recall table 2.1 for hydrogen-like atom orbitals (few slides back):

Orbital	n	l	m	$R_{n,l}(r)^a$	$Y_{l,m}(\theta, \phi)^b$
$2p_z$	2	1	0	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/4\pi)^{1/2} \cos \theta$

$$\rho = Zr/a_0$$

$$\zeta(\text{H}) = 1/2a_0 = 0.945 \text{ \AA}^{-1}$$

Also used as orbitals: Gaussian functions (easy to treat mathematically)

A linear combination of several Gaussians can replace the Slater-type orbital

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Molecular orbitals

Example: 3-methylindole (is a good model of the sidechain of tryptophan.)

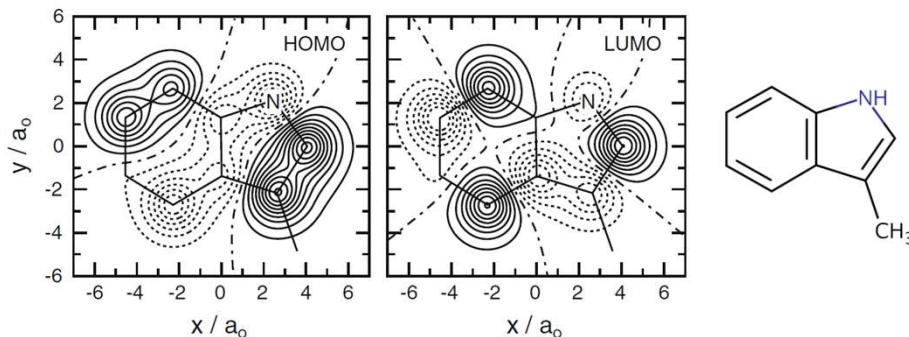


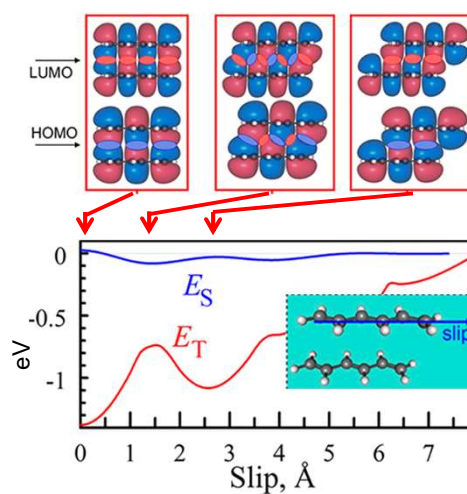
Fig. 2.8 Contour plots of the wavefunction amplitudes for the highest occupied molecular orbital (*HOMO*) and lowest unoccupied molecular orbital (*LUMO*) of 3-methylindole. Positive amplitudes are indicated by *solid lines*, negative amplitudes by *dotted lines*, and zero by *dot-dashed lines*. The plane of the map is parallel to the plane of the indole ring and is above the ring by a_0 as in Fig. 2.7, panels D and E. The contour intervals for the amplitude are $0.05a_0^{3/2}$. Small contributions from the carbon and hydrogen atoms of the methyl group are neglected. The *straight black lines* indicate the carbon and nitrogen skeleton of the molecule. The atomic coefficients for the molecular orbitals were obtained as described by Callis [37–39]. Slater-type atomic orbitals (Eq. 2.40) with $\zeta = 3.071/\text{\AA}$ ($1.625/a_0$) and $3.685/\text{\AA}$ ($1.949/a_0$) were used to represent C and N, respectively

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Molecular orbitals

Example:
two interacting hexane molecules
(Hartzler, Slipchenko, Savikhin 2018)

E_S and E_T are the binding energies of the dimer in the singlet ground state and in the lowest excited triplet state, respectively.



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Molecular orbitals

Multiple electrons and nuclei

The wave function for a system comprised of several independent components often can be approximated as a product of the wave functions of the components.

→ a wave function for a molecule can be written, to a first approximation, as a product of electronic and nuclear wave functions:

$$\Psi(r, R) \approx \psi(r)\chi(R)$$

where r and R are, respectively, the electronic and nuclear coordinates.

In this approximation, the total energy of the system is simply the sum of the energies of the electronic and nuclear wave functions. Similarly, a wave function for a molecule with N electrons can be approximated as a product of N one-electron wave functions.

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Large system orbitals

- Linear combinations of molecular wave functions can be used to generate approximate wave functions for systems containing more than one molecule.

For example, a wave function representing an excited state of an oligonucleotide can be described as a linear combination of wavefunctions for the excited states of the individual nucleotides.

In this approximation:

A thermodynamic change of state of a system in which the system exchanges energy with its surroundings by virtue of a temperature difference between them.

- Basis functions are *diabatic*, as they are solutions to Hamiltonians of individual molecules and do not account for interactions between different molecules.

The full wave function is then approximated as a superposition of diabatic basis wave functions

$$\psi_k = \sum_{n=1}^N C_n^k \psi_n^b$$

The Schrödinger equation for the full Hamiltonian is solved numerically: $\hat{H}\psi_k = E_k\psi_k$

The energies are then $E_k \approx \sum_{n=1}^N |C_n^k|^2 E_n$

Where E_n is the energy of basis wave function for noninteracting molecular Hamiltonian $\hat{H}_n^b \psi_n^b = E_n \psi_n^b$

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Large system orbitals

Solving Schrödinger equation for a system: matrix representation

Wavefunction:

$$\psi_k = \sum_{n=1}^N C_n^k \psi_n^b \quad \longrightarrow \quad \psi_k = \begin{bmatrix} C_1^k \\ C_2^k \\ \vdots \\ C_n^k \end{bmatrix}$$

Hamiltonian:

$$\hat{H} = \sum_{i=1}^N \hat{H}_{ii} + \sum_{i<j=1}^N \hat{H}_{ij} \quad \longrightarrow \quad \mathbf{H} = \begin{bmatrix} H_{11} & H_{12} & \cdots & H_{1n} \\ H_{21} & H_{22} & \cdots & H_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n1} & H_{n2} & \cdots & H_{nn} \end{bmatrix}$$

↑ interactions
For each molecule

Each element: $H_{ik} = \langle \psi_i | \hat{H} | \psi_k \rangle$

Integral over coordinates
of two electrons, r_i and r_k

Solve Schrödinger in matrix form: $\mathbf{H}\psi_k = E_k\psi_k$

$$= \int \psi_i^* \mathbf{H} \psi_k d\vec{r}_i d\vec{r}_j$$

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Large system orbitals

Solving Schrödinger equation for a system: matrix representation

Solve Schrödinger in matrix form:

$$\mathbf{H}\psi_k = E_k\psi_k$$

$$\begin{bmatrix} H_{11} & H_{12} & \cdots & H_{1n} \\ H_{21} & H_{22} & \cdots & H_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n1} & H_{n2} & \cdots & H_{nn} \end{bmatrix} \begin{bmatrix} C_1^k \\ C_2^k \\ \vdots \\ C_n^k \end{bmatrix} = E_k \begin{bmatrix} C_1^k \\ C_2^k \\ \vdots \\ C_n^k \end{bmatrix}$$

$$\psi_k = \sum_{n=1}^N C_n^k \psi_n^b$$

$$\hat{H} = \sum_{i=1}^N \hat{H}_{ii} + \sum_{i<j=1}^N \hat{H}_{ij}$$

$$H_{ik} = \langle \psi_i | \hat{H} | \psi_k \rangle$$

$$C_1^k H_{11} + C_2^k H_{12} + C_3^k H_{13} + \cdots + C_n^k H_{1n} = C_1^k E_k$$

$$C_1^k H_{21} + C_2^k H_{22} + C_3^k H_{23} + \cdots + C_n^k H_{2n} = C_2^k E_k$$

$$C_1^k H_{31} + C_2^k H_{32} + C_3^k H_{33} + \cdots + C_n^k H_{3n} = C_3^k E_k$$

⋮

$$C_1^k H_{n1} + C_2^k H_{n2} + C_3^k H_{n3} + \cdots + C_n^k H_{nn} = C_n^k E_k$$

**ERRORS
IN THE BOOK!!!**

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Large system orbitals

Solve system of equations → get C coefficients and energies

$$\psi_k = \sum_{n=1}^N C_n^k \psi_n^b$$

$$C_1^k H_{11} + C_2^k H_{12} + C_3^k H_{13} + \cdots + C_n^k H_{1n} = C_1^k E_k$$

$$C_1^k H_{21} + C_2^k H_{22} + C_3^k H_{23} + \cdots + C_n^k H_{2n} = C_2^k E_k$$

$$C_1^k H_{31} + C_2^k H_{32} + C_3^k H_{33} + \cdots + C_n^k H_{3n} = C_3^k E_k$$

⋮

$$C_1^k H_{1n} + C_2^k H_{2n} + C_3^k H_{3n} + \cdots + C_n^k H_{nn} = C_n^k E_k$$

This is an *eigenvalue* problem: the solution is equivalent to finding eigenvalues (i.e., energies E_k) and respective *eigenvectors* (C_n^k) of matrix **H**.

This is a standard mathematical procedure, and matrices with sizes exceeding 10,000x10,000 can be readily *diagonalized* using even common PC's.

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Spin wave functions

Spin: two quantum numbers

$$s = \frac{1}{2} \rightarrow \text{angular momentum magnitude } S = \hbar \sqrt{s(s+1)} = \hbar \sqrt{3} / 2$$

$$m_s = -s, \dots, s = -\frac{1}{2}, +\frac{1}{2} \rightarrow \text{projection to axis is } S_z = \hbar s = \pm \hbar / 2$$

Two spin functions:

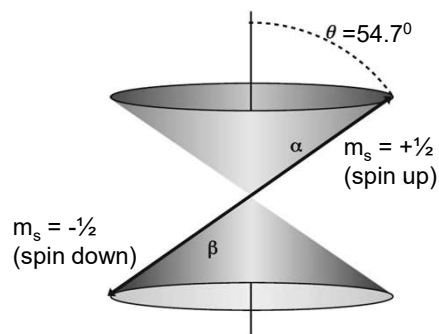
$$|\alpha\rangle \equiv \left| \uparrow \right\rangle - \text{spin up}$$

$$|\beta\rangle \equiv \left| \downarrow \right\rangle - \text{spin down}$$

Orthonormality:

$$\langle \alpha | \alpha \rangle = \langle \beta | \beta \rangle = 1$$

$$\langle \alpha | \beta \rangle = \langle \beta | \alpha \rangle = 0$$



Combine spatial part (ψ) and spin (χ): $\Psi = \psi(\vec{r}) \chi$

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Spin wave functions: HOMO

Two electrons:

Total $S = +1, 0$ (parallel or antiparallel)

Total $M_S = -S, \dots, S$

For most organic molecules HOMO contains 2 electrons with antiparallel spins
Since electrons are indistinguishable, wave function must be a linear combination of both spins

$$\Psi_a = [\psi_h(1)\psi_h(2)] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

Here ψ are the spatial wavefunctions for electron (1) and (2), which depend on the coordinates of the first and second electron, respectively
 $\alpha(1)\beta(2)$ means that electron 1 has spin up and electron 2 has spin down.

Note: wave function changes sign when electrons are swapped

Wave functions of all multielectron systems change sign when any two electrons are swapped!

→ *Wave function is antisymmetric*

→ *Pauli exclusion principle: each given spatial wave function can hold no more than two electrons*

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Spin wave functions: singlet LUMO

$$\text{HOMO: } \Psi_a = [\psi_h(1)\psi_h(2)] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

What happens when we excite one electron to a higher orbital?

Usually, the spin part of the wave function remains the same as for the ground state, so that both S and M_S remain zero.

The spatial part is more complicated:

$${}^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1) \right] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

Here ψ_h is spatial function for HOMO, and ψ_l for LUMO

Why there is a plus sign for spatial functions?

Note:

This LUMO is *singlet* state since spatial functions are symmetric and there is only one possible combination of spin functions

The ground state is also singlet

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Spin wave functions: Pauli principle

$$\Psi_a = [\psi_h(1)\psi_h(2)] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

Consider combination:

$$\Psi_a = \frac{1}{\sqrt{2}} [\psi_h(1)\sigma(1) \cdot \psi_h(2)\sigma(2) \pm \psi_h(2)\sigma(2) \cdot \psi_h(1)\sigma(1)]$$

↑
= $\alpha(1)$ or $\beta(1)$, spin state

Suppose both electrons are in the same spatial and spin state:

$$\Psi_a = \frac{1}{\sqrt{2}} [\psi_h(1)\alpha(1) \cdot \psi_h(2)\alpha(2) \pm \psi_h(2)\alpha(2) \cdot \psi_h(1)\alpha(1)]$$

What is wave function if we chose '-' sign?

What is wave function if we chose '+' sign?

Antisymmetric ('-') wave function changes sign when two particles swap

Fermions: using antisymmetric wave function satisfies Pauli principle

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Spin wave functions: triplet LUMO

$$\text{HOMO: } \Psi_a = [\psi_h(1)\psi_h(2)] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

$$\text{LUMO: } {}^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1) \right] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$$

What happens when we chose antisymmetric combination of wave functions?
For LUMO?

$${}^3\Psi_b^{+1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] [\alpha(1)\alpha(2)]$$

$${}^3\Psi_b^0 = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] \left[2^{-1/2}\alpha(1)\beta(2) + 2^{-1/2}\alpha(2)\beta(1) \right]$$

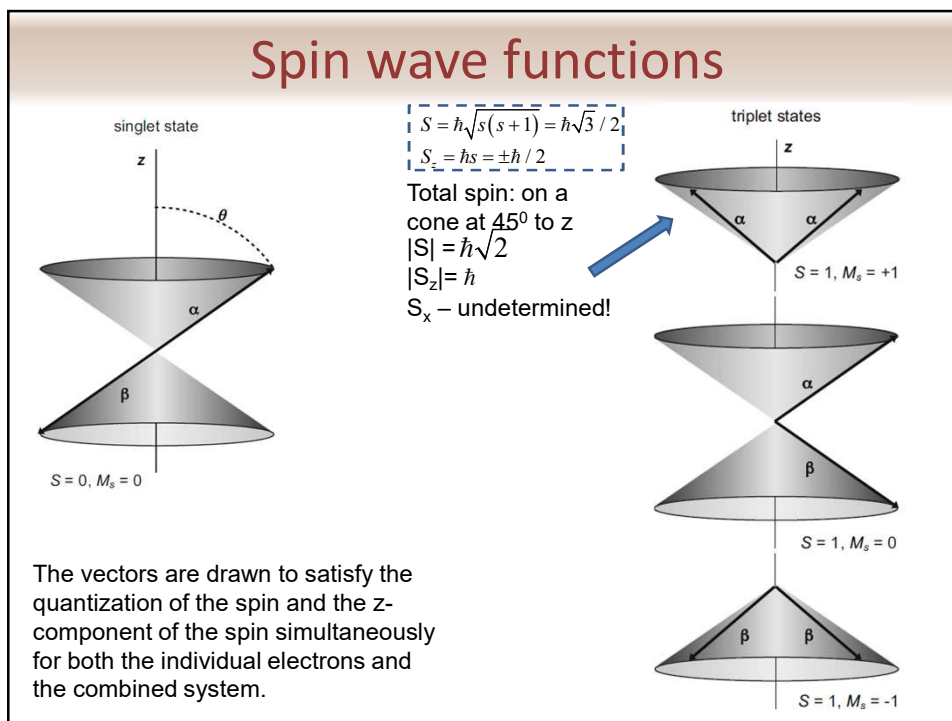
$${}^3\Psi_b^{-1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] [\beta(1)\beta(2)]$$

Triplet state: three possibilities for $M_s = -1, 0, +1$

Note: all wave functions must be antisymmetric, so S=0 will not work

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Spin wave functions



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Spin wave functions: energies

Singlet: ${}^1\Psi_b = \left[2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1) \right] \left[2^{-1/2}\alpha(1)\beta(2) - 2^{-1/2}\alpha(2)\beta(1) \right]$

Triplet: ${}^3\Psi_b^{+1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] [\alpha(1)\alpha(2)]$

${}^3\Psi_b^0 = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] \left[2^{-1/2}\alpha(1)\beta(2) + 2^{-1/2}\alpha(2)\beta(1) \right]$

${}^3\Psi_b^{-1} = \left[2^{-1/2}\psi_h(1)\psi_l(2) - 2^{-1/2}\psi_h(2)\psi_l(1) \right] [\beta(1)\beta(2)]$

Singlet-triplet splitting:

- Triplet state ${}^3\Psi_b^0$ has *lower* energy than singlet state ${}^1\Psi_b$.

This is a consequence of different *spatial* wavefunctions $\rightarrow$ the motion of two electrons is correlated in a way that tends to keep electrons further away from each other in triplet, decreasing repulsive interaction (*Hundt's rule*).

The difference is called *singlet-triplet splitting* and is equal to $2K_{hl}$, where

$$K_{hl} = \left\langle \psi_l(1)\psi_h(2) \left| \frac{e^2}{r_{12}} \right| \psi_h(1)\psi_l(2) \right\rangle. \quad \leftarrow \text{Exchange integral } (>0)$$

Special case: when two orbitals are equivalent in energy and can be transformed into each other by a symmetry operation (such as rotation) $\rightarrow$ triplet can be lower in energy, like in O_2 (triplet is ground state, singlet is excited).

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